CONSTRUCTIVE ARTIN–REES, AND THE ACL CERTIFICATE FROM IT

Status. This note states the constructive Artin–Rees theorem as it is checked by Unit in `unit/sag/artin-rees`, and explains how the ZMT certificate of ACL [ACL, §4] follows from it. Proofs of the theorem itself are in `artin-rees.tex` (§§1–5); numbers "AR n" refer to that note. Every statement below is checked by Unit, except where marked *(paper)*.

1. The constructive Artin–Rees theorem

Rings are commutative, unital and sets. "Merely" means: under propositional truncation.

Definition 1 (AR 1)**.** A ring R is *CFN* if

(D) for $u_1, \dots, u_k, v \in R$ it is decidable whether $v \in \operatorname{span}(u)$;
(C) for $u_1, \dots, u_k \in R$ the module $\operatorname{Syz}(u)$ is merely generated by finitely many syzygies;
(N) R is merely the image of a presentation by codes that is noetherian in the sense of Coquand–Persson [CP].

CFN is a proposition. Discrete fields, $\mathbb{Z}$, $R[X]$, quotients by finitely generated ideals and localisations $R[1/f]$ of CFN rings are CFN (AR 10, 12, 13). For example $A = k[b][a]$, with k a discrete field, is CFN. The class is therefore large enough for finitely presented algebras over $\mathbb{Z}$ or over a discrete field. No choice principle and no excluded middle are used.

Powers of $I = (a, b)$ acting on a submodule P are defined inductively: $I^0 P = P$, and $x \in I^{m+1}P$ if $x = ax_1 + bx_2$ with $x_1, x_2 \in I^m P$. An element of $I^m P$ is thus a binary tree of depth m whose leaves lie in P.

Theorem 2 (Artin–Rees, AR 15)**.** *Let R be CFN, $a, b \in R$, $I = (a, b)$, and $N = \operatorname{span}(n_1, \dots, n_s) \subseteq R^r$. There merely is $c \in \mathbb{N}$ such that for every m and every $x \in R^r$*

$$x \in I^{m+c}R^r \cap N \implies x \in I^m\big(I^c R^r \cap N\big).$$

The converse inclusion is immediate. The quantifiers come in the order $\exists c\, \forall m\, \forall x$: one bound serves for every m. This uniformity is what the application needs, since m is chosen after c.

Idea of the proof. In the bigraded ring $R' = R[U][V]$ (CFN by AR 10 applied twice), consider the module $Q^+ \subseteq R'^r$ generated by the n_i and by $(U - aY)e_k$, $(V - bY)e_k$. By the degree corollary (AR 9), $Q^+ \cap R'^r$ is merely generated by finitely many $G_1, \dots, G_p$. The bound c is the sum of their total degrees. A "good" element splits in degree $m + c + 1$ into two halves of degree $m + c$, and this gives the step $I^{m+c+1} \cap N \subseteq I(I^{m+c} \cap N)$. The bound comes from noetherianity through (N); (D) and (C) make the generators G_i computable.

Remark 3. In Unit this is `ArtinRees.Over.artin-rees : Trunc[Bound]`, where `Bound` is the record of c and the implication above. The statement for a finitely presented M and a finitely presented submodule reduces to this one (*paper*, AR §5: write $M = R^r/K$ and take the preimage of N, which contains K). The principal case $I = (f)$, $M = R$, used for flatness rather than ACL, is proved separately (AR 14, `ArtinReesPrincipal`), for the uniform Kill lemma (AR 27).

Corollary 4 (coefficients, AR 26)**.** *Under the hypotheses of Theorem 2, if $x \in I^{m+c}R^r \cap N$, then merely $x = \sum_j \lambda_j n_j$ with every $\lambda_j \in I^m$.*

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Proof. The theorem gives a tree of depth m whose leaves lie in N. Each leaf is a combination of the n_j, and at a node $ax_1 + bx_2$ the coefficients combine to $a\lambda_{1,j} + b\lambda_{2,j} \in I^{m+1}$. (`PowSpan.Free.coefficients`.) □

This is the form in which Artin–Rees is used below. A combination that lies *deep* in I can be rewritten with coefficients that lie deep in I.

2. The ACL statement

Let A be a gcd domain (in the sense of `sag/gcd`) that is CFN, for instance $A = k[b][a]$. Let $a, b \in A$, $I = (a, b)$, and

$$p = x + af + bg, \qquad q = y + au + bv \qquad (f, g, u, v \in A[x, y]).$$

Let $P, Q \in A[X, Y, Z]$ be their homogenisations, of degrees d, e. Let L be any A-algebra with elements x, y satisfying $p(x, y) = q(x, y) = 0$.

Theorem 5 (ZMT certificate, ACL §4)**.** *There merely are $s, r, t \in L$ with*

$$s = 1 + ar + bt,$$

such that s, sx and sy are each roots of a monic polynomial over A.

Geometrically: the fibre of $\operatorname{Spec} L \to \operatorname{Spec} A$ over $a = b = 0$ is the single point $x = y = 0$, and the theorem gives an element $s \equiv 1$ modulo I such that $L[1/s]$ is finite over A there. This is the constructive content of Zariski's Main Theorem in this situation. In Unit the statement is `Certificate` of `ACLZMTCong`, and the proof for a CFN gcd domain is `ACLCertificateGeneral.certificate`. Its instance for $k[b][a]$ is `ACLCertificateK`.

3. From Artin–Rees to the certificate

The proof works in an explicit Čech model of the twisted structure sheaves of the projective closure $W = V(P, Q) \subseteq \mathbb{P}^2_A$. Here $K^j(n)$ is the module of j-cochains of degree n for the three standard charts, with coefficients Laurent polynomials, and

$$N^j(n) = P\,K^j(n - d) + Q\,K^j(n - e)$$

is the module of cochains divisible by the equations. Let res be restriction to the chart Z, followed by evaluation at (x, y); it sends N^0 to 0. There is a fixed 0-cochain $z^* \in K^0(-1)$ (the cochain Z^{-1}) with $\operatorname{res}(Zz^*) = 1$, but z^* is *not* closed modulo N^1. The proof has five steps; only step 3 uses Artin–Rees.

Step 1: approximations of z^* (data (F1)). Iterating $Z^{-1} = E\,Z^{-1} + P\,u$, with E a combination of a and b, gives the following for every $n \geq 1$: vectors $\rho_n \in A^r$ with coordinates in I^{n-1}, and $c_n \in I^n K^0(-1)$, with

$$(\beta_n) \qquad \delta_0 z^* \equiv G\rho_n + \delta_0 c_n \pmod{N^1}.$$

Here $G\rho = \sum_t \rho_t m_t$ is a combination of fixed generators m_t of the closed 1-cochains modulo coboundaries (`CechF1`, `FixedCochain`). In words, the obstruction to closing z^* is $G\rho_n$, a combination with coefficients in I^{n-1}.

Step 2: the module of liftable obstructions is finitely generated. Let

$$N = \{\rho \in A^r \mid \exists v \in K^0(-1),\ G\rho - \delta_0 v \in N^1\}.$$

By (β_n), $\rho_n \in N$, with $v = z^* - c_n$. The gcd data of P, Q (the common factor D_0, cofactors P', Q', and the conditions (S0) and (S1) of `HomLambda.Data`) describe N as a projection of the kernel

of a finite matrix $\mathcal{M}$ over A (AR 29, AR 30). Coherence of A then gives merely $u_1, \dots, u_p$ and $b_1, \dots, b_p \in K^0(-1)$ with

$$N = \operatorname{span}(u_1, \dots, u_p), \qquad Gu_j - \delta_0 b_j \in N^1$$

(`ACLRelations.corN`, AR 31).

Step 3: the correction, by Artin–Rees. Let c be an Artin–Rees bound for $N \subseteq A^r$ (Theorem 2). Fix m and let $n = c + m + 1$. Then $\rho_n \in I^{m+c}A^r \cap N$, so Corollary 4 gives $\rho_n = \sum_j \lambda_j u_j$ with $\lambda_j \in I^m$. Put

$$w = c_n + \sum_j \lambda_j b_j \in I^m K^0(-1), \qquad h = z^* - w.$$

Modulo N^1, $\delta_0 w \equiv \delta_0 c_n + \sum_j \lambda_j G u_j = \delta_0 c_n + G\rho_n \equiv \delta_0 z^*$. So

$$\delta_0 h \in N^1, \qquad h - z^* \in I^m K^0(-1)$$

(`ACLCorrection.correction`). With $m = 1$: $h = z^* + am_1 + bm_2$ is closed modulo N^1.

This is where the uniformity of the bound matters. Step 2 alone gives $\rho_n = \sum_j \mu_j u_j$ with arbitrary coefficients μ_j; then $\sum_j \mu_j b_j$ corrects $\delta_0 z^*$, but the correction is no longer small in I. Artin–Rees turns "ρ_n is deep in I and lies in N" into "ρ_n is a combination of generators of N with coefficients deep in I". That is exactly what keeps w in I^m.

Step 4: the element s. Multiplication by X, Y, Z sends h to 0-cochains Xh, Yh, Zh of degree 0, which are still closed modulo N^1. Restriction gives

$$s = \operatorname{res}(Zh) = \operatorname{res}(Zz^*) + a\operatorname{res}(Zm_1) + b\operatorname{res}(Zm_2) = 1 + ar + bt,$$

$$\operatorname{res}(Xh) = sx, \qquad \operatorname{res}(Yh) = sy$$

(`ACLCertificate.From`).

Step 5: integrality. Closed 0-cochains of degree 0 modulo N^0 form a ring, and modulo coboundaries they form a finitely generated A-module: data (F2) give coordinates $\kappa :$ closed sections $\to A^{s'}$ (`CechSections`). For a closed section γ, apply the dependence lemma (AR 32: in a module with (D) and the principle of free lists, a sequence f_n merely has some $f_N \in \operatorname{span}(f_0, \dots, f_{N-1})$) to $f_n = \kappa(\gamma^n)$ in $A^{s'}$, which satisfies both hypotheses because A is CFN (AR 4). This gives $\gamma^N - \sum_{i<N} x_i \gamma^i \in N^0$. Restricting, $\operatorname{res}(\gamma)$ is a root of a monic polynomial over A (`SeqDependence`, `ACLCertificate.Walk`). Applied to $\gamma = Zh, Xh, Yh$ this gives Theorem 5.

Each step produces a proposition from merely given data (bound, generators, coefficients, dependence). These are eliminated one after the other into the truncated conclusion, so no choice is needed.

4. Where noetherianity enters

Noetherianity (N) of A enters twice, both times through CFN:

- in step 3, through Artin–Rees (Theorem 2: degree corollary for $R' = A[U][V]$);
- in step 5, through the principle of free lists on $A^{s'}$.

Before this work, both steps used a coded noetherian presentation of A itself (`ReesLifting`, `GradedDependence`). The certificate now depends only on "A is a CFN gcd domain".

References

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[2] T. Coquand, H. Persson, Gröbner bases in type theory, TYPES'98, LNCS 1657 (1999).
[3] W. Ruitenburg, Constructions of finitely generated submodules of constructively noetherian modules, *Compositio Math.* 62 (1987), 47–52.